Sarah Brown-Schmidt (Vanderbilt Univ.) and Melissa Duff (Vanderbilt Univ. Medical Center)
sarah.brown-schmidt@vanderbilt.edu

The impact of traumatic brain injury on maintenance of referential uncertainty

Introduction: Spoken language understanding can be modeled as a process where multiple lexical representations are simultaneous activated proportionate to the evidence for them[1,2]. The graded activation is maintained over time[3], supporting future adjustments to the ongoing interpretation[4,5]. When processing spoken discourse, graded activation of candidate *referents* is similarly maintained over time, with acoustically-induced uncertainty (e.g., a pronoun manipulated to be ambiguous between *he* and *she*) probabilistically guiding online interpretation, and the ability to recover from initial misinterpretation up to 35 syllables later[6,7]. While simultaneous maintenance of probabilistic representations of the discourse over long delays may be typical of language processing in healthy adults, unknown is how these processes manifest in the many language users for whom age or cognitive differences change core aspects of language processing. As a case in point, traumatic brain injury (TBI) affects ~ 1.7 million Americans annually and is a leading cause of long-term disability[8]. TBI is associated with a range of cognitive-communicative deficits that interfere with everyday communication[9-10], but the underlying mechanisms of impairment are understudied. We examine the impact of TBI on the online processing of gradient acoustic information in spoken discourse.

Methods: Participants (Ps) were individuals with moderate-severe TBI in the chronic phase (6+ months post-injury), and demographically matched non-injured comparison Ps (NC). Using a version of a previously-vetted paradigm[6], eye-tracked Ps viewed a scene (**Fig1**) while listening to narratives that introduced two different-gender characters, and then referred to one with an acoustically manipulated pronoun that varied on a 7-step continuum from *he:she*. The shape word disambiguated the intended referent, allowing for recovery of the intended meaning. After each story, Ps clicked a button to indicate if the story matched the picture. We manipulate the acoustics of the pronoun (~he -> ~she), and whether the pronoun disambiguates to the 1st or 2nd mentioned of the two referents in the story. We analyze target gaze during interpretation of the pronoun and final interpretation of the story.

Results: Analysis of the gaze data (40 TBI; 41 NC, **Fig2**) used a dynamic *glmm* (dGLMM[11]). The TBI group was treated as the reference level in the analysis which revealed a significant effect of referent order due to more target fixations in the TBI group when the referent was mentioned 1st in the story ($b = .08$, $p = .005$), consistent with findings of a processing benefit for 1st-mentioned referents[12]. A *decrease* in target fixations the further the acoustic realization of the pronoun was to its canonical form ($b = -1.09$, $p < .0001$) reveals sensitivity in the TBI group to the gradient acoustic signal. An effect of group was due to significantly fewer target fixations in the TBI group ($b = .26$, $p = .006$), consistent with general processing difficulty. An interaction of the acoustic effect with participant group ($p<.0001$) was due to a significantly stronger acoustic effect in the NC group ($b = -1.39$, $p<.0001$). Analysis of final interpretation likewise revealed an interaction of the acoustic effect with group, such that participants with TBI were less sensitive to the acoustic manipulation ($b = -1.05$, $p<.0001$) than NC participants ($b = -1.71$, $p<0001$).

Discussion: Moderate-severe traumatic brain injury is common, and confers cognitive-communicative deficits that interfere with everyday communication, however little is known about how TBI impairs language use. Here we show sensitivity in TBI to both discourse cues associated with the prominence of candidate referents, and gradient acoustic cues to the intended lexical item (i.e., *he* vs. *she*). Yet, participants with TBI were overall less likely to fixate the target, and were less sensitive to these acoustic cues both online and after hearing disambiguating information at the end of the story, consistent with prior evidence of broad disruptions to language processing in TBI[9-10]. We speculate some aspects of cognitive-communication disorders may emerge not from an inability to perceive and use individual cues (such as individual acoustic features), but rather from disruptions in managing and integrating multiple cues (acoustics, discourse, visual) in complex and dynamic interactions.

Figure 1. Example display (note actual characters were shown to participants). Example narrative: *Daisy is playing the cello for Donald, as the telephone is ringing.* ***~S/he's*** *standing in a blue* <u>*square/triangle*</u> *but it doesn't look like anyone is going to answer the phone.*

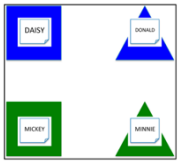
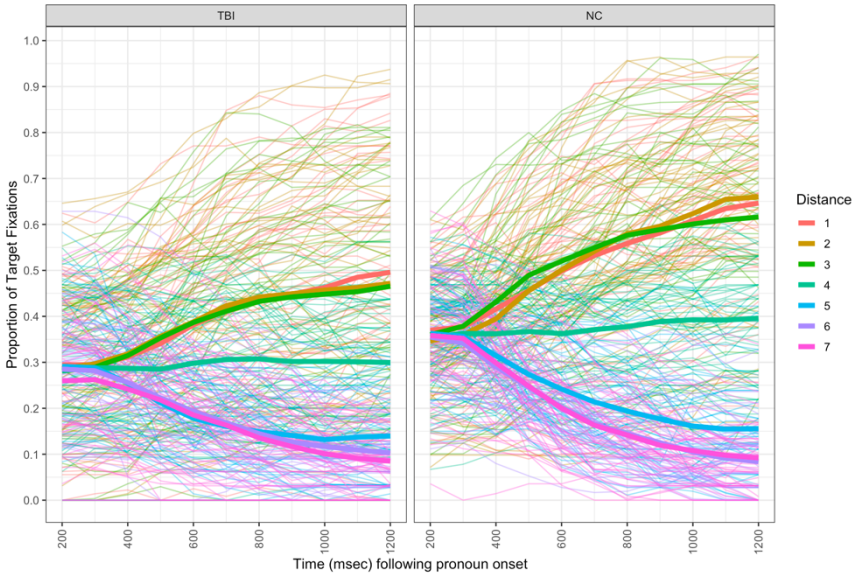


Figure 2. Fixations to the target following pronoun onset for individuals with TBI and non-injured comparison participants (NC). Colors denote the degree to which the pronoun acoustics differed from the intended referent (1 = close; 7 = far).



References

1. Allopenna, P. D., Magnuson, J. S., & Tanenhaus, M. K. (1998). Tracking the time course of spoken word recognition using eye movements: Evidence for continuous mapping models. *JML*, 38, 419–439.
2. McClelland, J., Mirman, D., & Holt, L. (2006). Are there interactive processes in speech perception? Trends in Cognitive Sciences, 10(8), 363–369. doi:10.1016/j.tics.2006.06.007
3. Dahan, D. (2010). The time course of interpretation in speech comprehension. Current Directions in Psychological Science, 19, 121–126. doi:10.1177/0963721410364726
4. McMurray, B., Tanenhaus, M. K., & Aslin, R. N. (2009). Within-category VOT affects recovery from "lexical" garden-paths: Evidence against phoneme-level inhibition. *JML*, 60, 65–91.
5. Connine, C. M., Blasko, D. G., & Hall, M. (1991). Effects of sub- sequent sentence context in auditory word recognition: Temporal and linguistic constraints. Journal of Memory and Language, 30, 234–250. doi:10.1016/0749-596X(91)90005-5
6. Brown-Schmidt, S., & Toscano, J. C. (2017). Gradient acoustic information induces long-lasting referential uncertainty in short discourses. *LCN*, *32*(10), 1211-1228.
7. Falandays, J. B., Brown-Schmidt, S., & Toscano, J. C. (2020). Long-lasting gradient activation of referents during spoken language processing. *JML*, *112*, 104088.
8. U.S. Department of Health and Human Services. (2022). Traumatic brain injury. National Institute of Neurological Disorders and Stroke. Retrieved October 4, 2023, from https://www.ninds.nih.gov/health-information/disorders/traumatic-brain-injury.
9. MacDonald, Martin, & Nath, (2017). *Trends in Canadian inpatient rehabilitation outcomes following traumatic brain injury*.
10. ASHA (2005). *Roles of speech-language pathologists in the identification, diagnosis, and treatment of individuals with cognitive-communication disorders: Position statement.*
11. Cho, S.-J., Brown-Schmidt, S., & Lee, W.-Y. (2018). Autoregressive generalized linear mixed effect models with crossed random effects: an application to intensive binary time-series eye tracking data. *Psychometrika.*
12. Arnold, J. E., Eisenband, J. G., Brown-Schmidt, S., & Trueswell, J. C. (2000). The rapid use of gender information: Evidence of the time course of pronoun resolution from eyetracking. *Cognition*, *76*(1), B13-B26.